



Experiment - 07

Determination of Voltage Regulation of Alternator by Direct loading

Aim- Determine of regulation of alternator by direct loading

APPARATUS REQUIRED:-

S.No	Name of Instruments	Specifications
1.	Ammeter	
2.	Voltmeter	
3.	Tachometer	

Theory:-

The Voltage Regulation of a Alternator is the rise in voltage at the terminals when the load is reduced from full load rated value to zero, speed and field current remaining constant. It depends upon the power factor of the load. For unity and lagging power factors, there is always a voltage drop with the increase of load, but for a certain leading power, the full load voltage regulation is zero. The voltage regulation is given by the equation shown below.

$$\text{Per Unit Voltage Regulation} \triangleq \frac{|E_a| - |V|}{|V|} \dots \dots \dots (1)$$

$$\text{Percentage Voltage Regulation} \triangleq \frac{|E_a| - |V|}{|V|} \dots \dots \dots (2)$$

Where, • $|E_a|$ is the magnitude of a generated voltage per phase • $|V|$ is the magnitude of rated terminal voltage per phase In this case, the terminal voltage is the same for both full load and no load conditions. At lower leading power factors, the voltage rises with the increase of load, and the regulation is negative.

The alternator runs at synchronous speed, and its terminal voltage is adjusted to its rated value V . The load is varied until the Ammeter and Wattmeter indicate the rated values at the given power factor. The load is removed, and the speed and the field excitation are kept constant. The value of the open circuit and no load voltage is recorded. It is also found from the percentage voltage regulation and is given by the equation shown below.

$$\% \text{ Voltage Regulation} = \frac{E_a - V}{V} \times 100\%$$

The method of direct loading is suitable only for small alternators of the power rating less than 5 KVA